RevTrace: Dual-Granularity Feedback Transactions for State-Certified Streaming ASR Serving

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Abstract

Streaming ASR systems often feed generated text into later rounds, turning a displayed revision into a write to future model state. Batching-induced output differences can therefore persist beyond one response. We formalize this feedback-state contamination channel and confirm it causally: among 21 legal paired interventions, 8 propagate with feedback and none do when the edge is removed. We introduce Dual-Granularity Feedback Transactions (DGFT), which decouple low-latency proposal and display from coarse canonical state-write authority. A revision enters future state only through a certificate-bearing, versioned transaction; same-granularity certification is the $\rho = 1$ case. We prove exclusion of uncertified ancestry, fork-free committed histories, and safety independent of granularity, and derive an authority-path cost decomposition. RevTrace implements DGFT and is evaluated on Qwen3-ASR 0.6B/1.7B and Whisper tiny/base/small.en on two RTX 3090 GPUs, plus an independent eight-V100 scale-out contract. On a matched Qwen workload, coarse authority improves throughput by 1.39×/1.28× over same-granularity configurations; across production contracts RevTrace reaches up to 6.88× wall-clock speedup and passes all 16 Whisper cells. We observe no output/state or transaction-install violation over 800 Qwen revisions and 1,048 Whisper commits; a 250,000-attempt fuzzer accepts all 10,000 valid commits and rejects all 240,000 mutations/replays. On the V100 contract, four proposal and four authority replicas sustain 147.1×/114.5× audio real-time throughput at c64 with display P99 1.115/1.471 s; c128 is retained as the measured 2 s-SLO boundary. These results establish state-write authority as a separate contract for safe, high-throughput revision serving; risk signals may schedule verification but never authorize installation.

Keywords: streaming ASR serving, LLM inference, optimistic concurrency control, state certification, transactional runtime

1. Introduction

Streaming speech services are increasingly built on *revision-generative* models: a Whisper-style decoder is conditioned on its own previously generated text [1, 2], and ASR-capable large language models such as Qwen3-ASR re-prompt themselves with rollback prefixes of their prior output [3]. In these systems the generated text is not merely displayed; it is written back into the state that conditions future generation. Mainstream throughput-oriented serving runtimes optimize request execution and output delivery but do not expose a distinct authorization boundary for model-generated projections before they become future feedback [4, 5, 6].

Batching is the standard mechanism by which serving systems buy throughput: independent sessions share one forward pass [4, 7, 8, 9]. But batched execution can change floating-point reduction order relative to singleton execution, so greedy outputs may diverge across batch configurations [10]. In terminal-output systems this difference remains local to one response. In revision-generative systems it is worse: once a perturbed output is written into feedback state, the divergence is inherited by every future revision. Section 2 makes this precise; our causal experiment confirms the prediction: with the feedback edge present, 8 of 21 legal state pairs propagate a batch-created divergence, and with the edge removed, none do; on the persistent pairs the measured revision amplification factor (RAF) reaches 153.

Prior mechanisms do not cover this failure mode. Streaming ASR confirmation policies such as LocalAgreement commit stable display prefixes but never constrain which outputs may become future-state parents [2]. Speculative decoding verifies draft tokens against a target distribution, not state-write authority [11, 12]. LLM-42 is the closest execution mechanism: it combines a nondeterministic fast path with verify–rollback for deterministic token generation, but it does not define which revision projection may become a parent of future multimodal state [13]. Numerical reproducibility work repairs arithmetic execution rather than feedback authorization [10]. Optimistic concurrency control supplies the closest analogy [14, 15], yet classical transactions commit reads and writes of stored data, whereas here the "write" is a model-generated projection whose legality must be certified by the serving contract itself.

This paper takes one position: *revision outputs are speculative writes to future model state, not merely user-visible responses*, and

$$\text{Display}(Y_t) \;\not\Rightarrow\; \text{Authorize}(\Phi(Y_t; H_t)).$$

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We introduce **Dual-Granularity Feedback Transactions** (DGFT): a fine proposal path serves low-latency display, while only a coarse canonical authority may install state projections, and only as a certificate-bearing transaction. DGFT is formalized in Section 3, where same-granularity certification appears as the degenerate $\rho = 1$ case, three transaction invariants are proved, and an authority-cost decomposition states the operating condition under which coarse authority outperforms same-granularity verification. **RevTrace** (Sections 4–5) implements the transaction runtime with a stratified, contract-scoped certification layer.

Contributions:

1. A problem characterization of feedback-state contamination in revision-generative serving, with a causal verification that the feedback edge is the propagation channel (Section 2, Section 6, H1).
2. DGFT, a dual-granularity transaction mechanism with provable invariants (uncertified-ancestry exclusion, fork-free committed-state linearity, and granularity-independent safety) and an authority-path cost decomposition tested against complete operating configurations (Section 3, H3).
3. RevTrace, a runtime with contract-scoped stratified certification: randomized audits and anytime-valid bounds on the same-granularity path, FP32 preflight invariance checks, exhaustive transaction validation, and independent quality references (Section 4, H2, H5).
4. A cross-family evaluation on two architecture families and five checkpoints (Qwen3-ASR 0.6B/1.7B; Whisper tiny/base/small.en) on two RTX 3090 GPUs: same-workload SG/Coarse/Hybrid ablation, production performance, per-family quality contracts, and boundary evidence; plus an independent 8×V100 Qwen scale-out matrix that separates the proposer and authority bottlenecks and locates the 2 s-SLO capacity boundary (Section 6, H1–H7).

2. Model and Problem Formulation

2.1. Revision-generative state machines

Definition 2.1 (Revision-generative serving round)**.** *A serving round t of a revision-generative system is a tuple (S_t, A_t, Y_t) where S_t is the committed feedback state entering the round, A_t is the exogenous input (audio frontier), and Y_t is the generated output. The state update is*

$$S_{t+1} = \mathcal{T}(S_t, A_t, Y_t), \tag{1}$$

where $\mathcal{T}$ is the model/adapter state-update function. The system is revision-generative *when $\mathcal{T}$ depends on Y_t, i.e. the output of round t participates in the state read by round $t{+}1$.*

Not every byte of Y_t enters future state. We therefore isolate the feedback sub-state from the exogenous and runtime conditions held fixed in a causal comparison.

Definition 2.2 (Feedback sub-state and state projection)**.** *Write $S_t = (S_t^x, H_t)$, where H_t is the* feedback sub-state *written from generated output and S_t^x denotes the exogenous/runtime state conditioned on in the comparison (model and decoder configuration, input schedule, and batch assignment). We require paired causal executions to hold S_t^x fixed; we do not claim that every cache or runtime buffer is invariant in an arbitrary deployment. The state projection $\Phi(Y_t; H_t)$ is the adapter map that turns the round-t output into the next feedback sub-state,*

$$H_{t+1} = \Phi(Y_t; H_t), \tag{2}$$

so that the full update of Eq. (1) *factors as*

$$S_{t+1} = \mathcal{G}(S_t^x, H_{t+1}, A_t), \tag{3}$$

with generated output affecting the compared future state through H_{t+1}. For Qwen3-ASR, H_t is the rollback prefix re-prompted at the next revision [3]; for Whisper with previous-text conditioning it is the conditioning prompt [1].

Ordinary batched serving makes an implicit assumption we name explicitly:

$$H_{t+1} = \Phi(Y_t^{\text{batch}}; H_t), \tag{4}$$

i.e. whatever the batched forward pass produced may write the feedback sub-state, with no authorization step between generation and state write. Equation (4) conflates "was generated" with "may write future state".

2.2. Display output versus state projection

The user-facing display function may surface output that carries no future-state authority: display-only suffixes, hold-back regions, and provisional hypotheses are shown but excluded from $\Phi(Y_t; H_t)$ by the adapter (Section 4). The paper's central invariant separates the two permissions:

$$\boxed{\text{Display}(Y_t) \;\not\Rightarrow\; \text{Authorize}(\Phi(Y_t; H_t))} \tag{5}$$

2.3. State-entry divergence and contamination

Let Y_t^f be the fine (batched, proposer) output and Y_t^a the canonical authority output for the same round.

Definition 2.3 (Output divergence)**.** *$\varepsilon_t = d_Y(Y_t^f, Y_t^a)$, with d_Y an output-space distance (mixed-unit edit distance in our contracts).*

Definition 2.4 (State-entry divergence)**.** *Let $P_t^f := \Phi(Y_t^f; H_t)$ and $P_t^a := \Phi(Y_t^a; H_t)$. State-entry divergence is $e_t = d_S(P_t^f, P_t^a)$, where d_S is the distance on feedback sub-states induced by the same mixed-unit metric.*

Definition 2.5 (Contamination event)**.** *Round t is a contamination event C_t iff the state projection of a fine proposal is installed into the feedback sub-state while $e_t > 0$:*

$$C_t \;:\Longleftrightarrow\; e_t > 0 \;\wedge\; \Phi(Y_t^f; H_t) \text{ installed into } H_{t+1}.$$

Definition 2.5 upgrades the problem statement from "outputs differ" to "a differing output acquired future-state write permission". Display-level instability literature observes the former; this paper targets the latter.

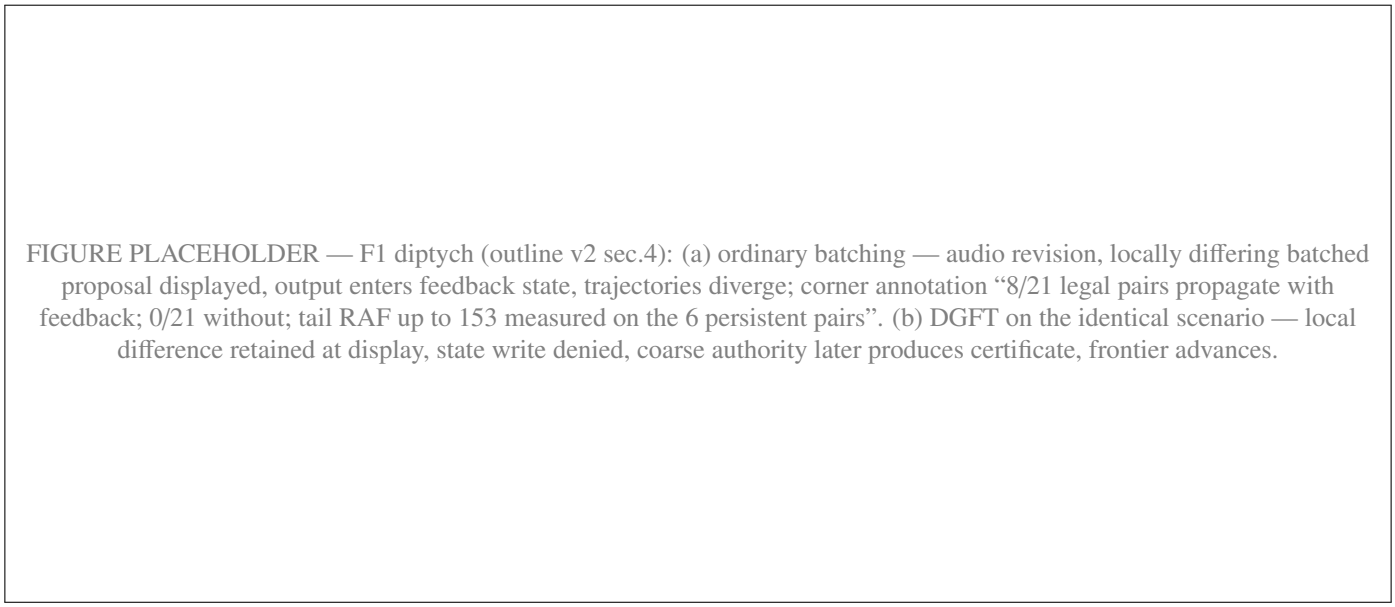


Figure 1: State-entry contamination under ordinary batching (a) and the same scenario under DGFT (b). DGFT does not require every provisional output to be identical; it prevents uncertified outputs from becoming future-state parents.

2.4. Feedback amplification

Let

$$D_t := d_H(H_t^f, H_t^a)$$

be the feedback-trajectory distance between the batched-served and canonical executions under identical future exogenous inputs and runtime assignments. This definition deliberately makes no claim that the distance between full runtime states is carried entirely by H_t.

Assumption 2.1 (Local sensitivity)**.** *In a neighborhood of the canonical trajectory, the state update has finite per-round sensitivity: there exist finite coefficients L_t and β_t such that the decomposition* (6) *below holds for trajectories inside that neighborhood. We make no global Lipschitz claim, do not assume L_t, β_t are known in advance, and use them only to identify propagation channels.*

Under Assumption 2.1, a local perturbation decomposition gives

$$D_{t+1} \;\le\; L_t\, D_t \;+\; \beta_t\, e_t, \tag{6}$$

where L_t measures propagation of pre-existing difference and β_t measures coupling of the round-t state-entry divergence into the next state. Equation (6) is a per-round decomposition with unknown coefficients, used only to identify channels.

Remark 2.2 (Causal prediction)*.* Removing the feedback edge sets $\beta_t \equiv 0$, so an isolated round-t state-entry perturbation can no longer propagate through that revision edge, provided subsequent exogenous inputs, batch assignments, and arithmetic realizations are held equal. This intervention does not rule out a fresh, independent divergence in a later round. Section 6 (H1) tests the scoped prediction.

Definition 2.6 (Revision amplification factor)**.** *For horizon K, the measured trajectory amplification factor is*

$$\mathrm{RAF}_{t,K} \;=\; \frac{D_{t+K}}{e_t + \epsilon_0},$$

an empirical indicator of how one state-entry divergence propagates through subsequent revisions. RAF is reported as a measured trajectory quantity, not as a theoretical bound.

2.5. Design requirements derived from the model

The model above forces four requirements on any serving mechanism that claims to certify feedback state:

R1 (Authority separation). Display permission and state-write permission are distinct grants; Eq. (5).

R2 (Parent ownership). A state projection commits only onto the parent version it was computed against; otherwise staleness is undetectable.

R3 (Frontier monotonicity). Committed audio frontiers advance monotonically, with no gap and no overlap, so that the committed history is a single sequence.

R4 (Evidence non-authority). Statistical evidence (margins, coverage, repetition signals) schedules verification; it never authorizes installation.

R1–R4 reappear as protocol properties of DGFT (Section 3), labels on the system diagram (Figure 2), and columns of the guarantee taxonomy (Table 1).

3. DGFT: Theory and Mechanism

3.1. Transactions, certificates, and commits

DGFT wraps every state projection in a transaction. Throughout this section, "committed state" means the committed feedback sub-state H of Section 2; the remaining serving state S^x is never written by generated output and therefore needs no certification.

Trust model.. The authority coordinator and versioned ledger are trusted runtime components; proposers and clients cannot mint or modify certificates. We assume collision resistance of h, atomic certificate consumption with the ledger compare-and-swap, and immutable session audio bytes once covered by a certificate. The threat model excludes a Byzantine authority or compromised ledger. A digest provides content binding, not standalone authentication; authenticity follows from issuance inside the protected coordinator.

The protocol theorems do not assume a deterministic proposer, greedy decoding, synchronous GPU kernels, or equality of proposer KV caches. The authority may use greedy, beam, or sampled decoding provided that the realized projection and the complete decoding/runtime configuration (including any randomness policy needed by the contract) are bound by χ_j. Asynchrony is allowed because the parent-version compare-and-swap serializes installation. The theorems certify provenance and installation legality, not semantic correctness of the authority output or repeatability of an independently rerun stochastic authority.

Definition 3.1 (DGFT transaction)**.** *A transaction is the object*

$$\tau_t = (\text{sid},\ e,\ p,\ v_{\text{parent}},\ [a_s, a_e),\ Y_t,\ \Phi(Y_t; H_t),\ \text{cert}_t),$$

comprising the session id, contract epoch e, policy p, the parent state version read, the audio frontier interval covered, the generated output, its state projection, and the certificate it carries (pending for fine proposals, issued for authority commits). The proposal granularity g_p (fine window) and the authority granularity g_a (coarse canonical segment) define the granularity ratio $\rho = g_a/g_p \geq 1$.

Definition 3.2 (Certificate)**.** *An authority certificate is a binding record*

$$\text{cert}_j = \text{Bind}(\text{sid}_j, e_j, p_j, v_j, [f_j, f_{j+1}), \\ h(A_{[f_j, f_{j+1})}), h(\Phi^a_j), \chi_j),$$

where $\Phi^a_j = \Phi(Y^a_j; H_{v_j})$ is the canonical state projection computed by the authority against parent version v_j, and $h(\cdot)$ is a content digest over immutable bytes. The configuration digest χ_j binds the model checkpoint, tokenizer, decoding parameters, projection adapter, numerical precision, and authority-runtime version. A certificate therefore binds exactly one canonical projection to exactly one commit tuple; it is issued only when the contract's authority checks pass, and it is single-use: consumed by the commit that presents it.

Definition 3.3 (Commit validity)**.** *An authority commit is a tuple*

$$\Gamma_j = (\text{sid}_j,\ e_j,\ p_j,\ v_j,\ [f_j, f_{j+1}),\ \Phi^a_j,\ \text{cert}_j).$$

Γ_j is valid*, written* Valid(Γ_j)*, iff (i) session, contract epoch, and policy match the live session fence; (ii) v_j equals the currently committed parent version; (iii) $[f_j, f_{j+1})$ continues the committed frontier without gap or overlap; and (iv) certificate binding: cert_j binds exactly the presented fields, the digest $h(\Phi^a_j)$ recomputes from the presented projection, and the certificate has not been consumed before. Only valid commits mutate committed state.*

Definitions 3.1–3.3 realize requirements R1–R4 (Section 2.5) structurally: display never produces a commit (R1), v_j enforced inside the certificate binding gives parent ownership (R2), $[f_j, f_{j+1})$ enforces frontier monotonicity (R3), and cert_j is issued only by authority checks, never by risk signals (R4). Table 1 previews how these properties separate DGFT from weaker protection regimes; the guarantee claims are evaluated in Section 6 (H2, H5).

Definition 3.4 (Committed provenance ancestry)**.** *A projection P is an ancestor of committed state H_j iff P labels an accepted ledger transition on the unique path $H_0 \to H_1 \to \cdots \to H_j$. Displayed proposals and rejected transactions are not ancestors.*

3.2. Theorem 1: uncertified-ancestry exclusion

Theorem 3.1 (Uncertified-Ancestry Exclusion)**.** *If committed state is updated exclusively through commits satisfying* Valid *(Definition 3.3), then no fine projection that is not bound by a valid certificate can become part of the ancestry of any future committed state in the sense of Definition 3.4.*

Proof. By induction over the commit sequence $\Gamma_1, \Gamma_2, \ldots$. Base: the initial committed state H_0 is contract-defined, so its ancestry contains no generated output. Step: assume committed state H_j has ancestry containing only certificate-bound projections. The next state H_{j+1} arises only from a valid Γ_{j+1}; by Definition 3.3(iv), commit application installs exactly the projection whose digest is bound by cert_{j+1}, since any other content fails the binding check and is rejected. A fine projection $\Phi(Y^f; H)$ that differs from the certified canonical projection Φ^a_{j+1} therefore cannot be installed by Γ_{j+1}; if a fine proposal happens to coincide bitwise with Φ^a_{j+1}, what enters state is the certified content, attributable to the authority's certificate rather than to the uncertified fine generation. Provisional display never mutates committed state by construction of the runtime (Section 4), completing the induction. □

Theorem 3.1 is deliberately architecture-agnostic: it depends on the commit protocol, not on margin accuracy, model determinism, FP16/FP32 agreement, or the choice of ASR family.

Table 1: Guarantee taxonomy. ✓: guaranteed within the stated scope; "display only" means that a policy stabilizes what the user sees without defining feedback-state write authority. Rows 2–4 isolate single-protection regimes; they are not strawmen but the actual components of existing practice. The last four columns are the certification strata of Section 4.4, each with a non-transferable scope (H5): *statistical risk bound* = anytime-valid UCB from randomized audits, SG adapter path only (Section 6.6(i)); *preflight invariance* = contract-scoped FP32 batch-repeat and batch-vs-singleton checks (Section 6.6(ii)), not applicable to the singleton SG authority; *exhaustive tx validation* = exhaustive online validation of certificates, parent versions, and frontier continuity (Section 6.3); *semantic reference* = independent per-family quality reference (Section 6.6(iv)). Margin and confidence signals schedule verification but authorize nothing (R4), so the margin-gate row carries no evidence column.

Regime	Display/state-write authority split	Feedback ancestry	Canonical authority	Stale rejection	Statistical risk bound	Preflight invariance	Exhaustive tx validation	Semantic reference
Unprotected FP16 batching	×	×	×	×	×	×	×	×
Owner/version check only	×	×	×	✓	×	×	×	×
LocalAgreement-style confirmation	display only	×	×	×	×	×	×	×
Margin/confidence gate	×	×	×	×	×	×	×	×
SG-DGFT ($\rho = 1$)	✓	✓	✓	✓	✓	×	✓	✓
Coarse/Hybrid DGFT ($\rho > 1$)	✓	✓	✓	✓	×	✓	✓	✓

3.3. Theorem 2: fork-free committed-state linearity

Theorem 3.2 (Fork-Free Committed-State Linearity)**.** *Assume commit application is atomic: the validity checks of Definition 3.3 and the state installation execute as one indivisible compare-and-swap against the committed state, and each certificate is consumed by at most one commit. Then, under parent-version checking (Definition 3.3(ii)) and frontier-continuity checking (Definition 3.3(iii)), the accepted commits of a session form a fork-free, gap-free, overlap-free linear history of feedback state.*

Proof. Atomicity closes the check-then-act race: two distinct commits cannot both observe the same committed state, both pass the checks, and both install. Parent-version checking then makes the accepted commits a tree of height at most one per version: two distinct commits cannot both satisfy $v_j = \text{version}(H)$ against the same committed state, so the history is fork-free. Single-use consumption excludes replay of an already-consumed certificate against a later parent. Frontier-continuity checking orders accepted intervals $[f_j, f_{j+1})$ into a sequence with $f_{j+1} = f'_j$ for consecutive commits, excluding gaps and overlaps. Hence the accepted commits are a single linear sequence. □

Remark 3.3 (Scope of the concurrency claim). Theorem 3.2 proves a serial accepted-commit history with the ledger compare-and-swap as its serialization point. It is not a claim of full Herlihy–Wing linearizability for every client invocation and response; such a claim would require a separate concurrent-object history and real-time-order specification [16].

Corollary 3.4 (Stale-descendant rejection)**.** *A revision computed against a stale parent, i.e. $v_{\text{parent}} \neq \text{version}(H_j)$, cannot legally commit. It fails Definition 3.3(ii) and is rejected without mutating state.*

3.4. Theorem 3: granularity-independent safety

Theorem 3.5 (Granularity-Independent Safety)**.** *For any granularity ratio $\rho \geq 1$, if every authority segment starts at the committed parent, covers a contiguous audio frontier, yields a legal state projection, and passes the certificate and version checks of Definition 3.3, then the conclusions of Theorems 3.1 and 3.2 hold.*

Proof. The proofs of Theorems 3.1 and 3.2 use only the certificate binding of Definition 3.2, the validity predicate of Definition 3.3, and atomic commit application; none of their steps depends on the size of $[f_j, f_{j+1})$ relative to the fine window. Varying ρ changes how much audio a single certificate covers, not which commits are admissible. □

Corollary 3.6 (Same-granularity certification as $\rho = 1$)**.** *Setting $\rho = 1$ makes proposal and authority granularities coincide: the directional coverage witness degenerates to the identity and fine replay degenerates into risk-triggered re-verification. The same-granularity (SG) configuration is therefore a reduction of DGFT under the same protocol, not a separate mechanism.*

Remark 3.7 (Canonical-path boundary). Corollary 3.6 identifies the protocol, not the arithmetic path: in our contracts SG runs an FP16 singleton authority while Coarse/Hybrid run an FP32 batched authority. These are distinct canonical paths; we claim protocol equivalence, never bitwise identity.

3.5. Authority-cost decomposition

The safety results above are silent about cost; the following analysis is a proposition with explicit assumptions, whose predictions we test (Section 6, H3). Let B be the number of concurrent sessions, q the SG verification fraction, r the Hybrid selective-replay fraction, C_s the singleton authority cost per covered region, $C_c(\rho, B)$ the batched coarse authority cost per segment, C_{commit} the transaction/commit overhead, and Q_s the queueing/fragmentation cost of singleton authority. Then

$$T^a_{\text{SG}} \approx q\, B\, \rho\, C_s \;+\; Q_s, \tag{7}$$

$$T^a_{\text{DGFT}} \approx C_c(\rho, B) \;+\; r\, B\, \rho\, C_s \;+\; C_{\text{commit}}, \tag{8}$$

and end-to-end service time is governed by the slower lane plus repair:

$$T_{\text{service}} \approx \max(T_p, T^a) + T_{\text{repair}}, \tag{9}$$

with proposer and authority running concurrently on separate GPUs.

Proposition 3.8 (Model-based operating condition)**.** *Let end-to-end service time be given by Eq.* (9)*. Coarse-DGFT improves service throughput over SG whenever*

$$\max(T_p,\ C_c(\rho, B) + rB\rho C_s + C_{\text{commit}}) + T^D_{\text{repair}} \;<\; \max(T_p,\ qB\rho C_s + Q_s) + T^S_{\text{repair}}.$$

A simplified sufficient condition, applicable when the authority lane is on or near the critical path in both configurations ($T^a \geq T_p$) and the repair costs are comparable, is

$$C_c(\rho, B) + rB\rho C_s + C_{\text{commit}} < qB\rho C_s + Q_s.$$

In particular, even a small verification fraction q leaves SG paying singleton invocation and queueing costs across all B sessions, while a batched coarse authority amortizes them into C_c.

Equations (7)–(8) motivate four empirical hypotheses rather than determine them without fitted costs: (P1) a burst of singleton authority work can create an SG tail even at 8–10% verification; (P2) a batched coarse authority can amortize invocation and queueing costs; (P3) Hybrid may trade replay fraction r for quality at a performance and energy cost; and (P4) the ordering can reverse across models because the complete authority paths—including precision and execution mode—have model-dependent costs. Section 6 tests these hypotheses on the same workload. Because SG and Coarse/Hybrid use different canonical arithmetic paths in our implementation (Remark 3.7), the comparison identifies integrated operating points, not a precision-independent causal effect of ρ alone.

3.6. Operating-point trade-offs

A deployment selects SG, Coarse, or Hybrid as configuration policy π, trading latency, energy, and quality subject to hard safety:

$$\max_{\pi} J(\pi) \quad \text{s.t.} \quad V(\pi) = 0,\ C(\pi) \leq B,$$

where J aggregates the deployment objective, V counts transaction violations, and $C(\pi) \leq B$ denotes the contract's display and quality caps. DGFT supplies the safety-constrained operating envelope; the three evaluated contracts instantiate three operating points. We do not solve this optimization online and do not claim any single operating point dominates: the measured ordering is workload- and model-dependent (Section 6, H3).

4. The RevTrace System

RevTrace implements DGFT as a transaction runtime: per-family adapters, a versioned ledger, and a stratified certification layer (Figure 2).

4.1. Architecture-specific adapters

Motivation. The safety proofs of Section 3 require only three interfaces; everything family-specific must stay out of the protocol. *Mechanism.* Each family provides (i) a state-projection adapter computing $\Phi(Y_t; H_t)$ (Qwen rollback prefix; Whisper previous-text prompt), (ii) an authority adapter executing the canonical path (FP16 singleton for SG; FP32 batched for Coarse/Hybrid), and (iii) an evidence adapter collecting risk signals (margins, directional coverage, repetition) for the certification layer. *Evidence hook.* H4 (Section 6) runs the identical runtime over both families through these adapters.

4.2. Certificate and versioned ledger

Motivation. R2/R3 require detecting staleness and frontier discontinuity at commit time. *Mechanism.* A compare-and-swap-style ledger stores committed state versions; every commit tuple Γ_j (Definition 3.3) is checked against session, contract epoch, policy, and state-version fences before installation, and each accepted commit consumes its certificate exactly once. *Evidence hook.* H2: exhaustive transaction validation reports every install attempt against these fences (320 Qwen events; 1,048 Whisper commits; zero violations).

4.3. Directional coverage and selective fine replay

Motivation. A coarse authority segment must be witnessed against the fine proposals it replaces; replaying everything would forfeit the point of coarse authority. *Mechanism.* The coverage witness records, in Latin and CJK units, which fine-region outputs the coarse certificate accounts for; uncovered regions are queued for fine replay. Hybrid adds risk-triggered replay at fraction r, trading cost for quality (Eq. (8)). *Evidence hook.* H3 ablates $r = 0$ (Coarse) against $r > 0$ (Hybrid) on the same workload.

4.4. Stratified certification

Four strata, each with a non-transferable scope (Figure 3): (i) *SG numerical risk*: randomized audits of the singleton authority with anytime-valid (UCB) bounds, scoped to the SG adapter path [17, 18]; (ii) *authority implementation*: contract-scoped FP32 preflight invariance checks (batch repeat, batch-vs-singleton), fail-closed; (iii) *transaction integrity*: exhaustive online validation of certificates, versions, and frontier continuity; (iv) *semantic quality*: independent per-family references. A risk signal from any stratum may schedule verification but never authorize installation (R4).

4.5. Online coordinator

The coordinator schedules the proposer and authority lanes on separate GPUs (Eq. (9)), invalidates stale descendants (Corollary 3.4), closes terminal frontiers at session end, and performs rollback/rebase when a commit supersedes displayed provisional state.

5. Implementation

RevTrace is implemented as a serving runtime for revision-generative ASR models. The Qwen rollback adapter maintains a growing audio buffer with prefix rollback and holdback semantics; the Whisper prompt adapter maintains previous-text conditioning state [3, 1]. The coarse authority executes in FP32 with a 4,096-token KV-cache bound per session; the one-time contract preflight (batch-repeat and batch-vs-singleton invariance) takes 10.7–11.1 s and is excluded from steady-state service timing, reported separately (Section 7). Sessions arrive online and are scheduled with causal arrival ordering; decoder outputs are transferred to the host via vectorized paths. Experiments use two NVIDIA RTX 3090 GPUs, one hosting the proposer lane and one the authority lane. Realizations rejected

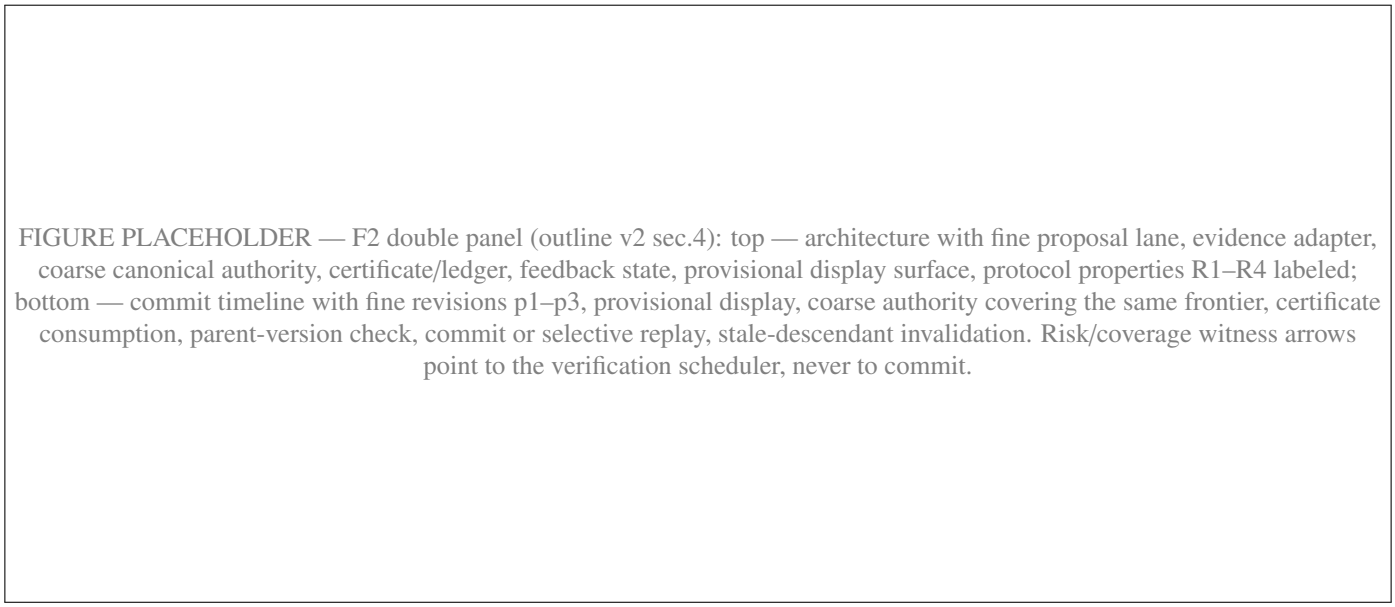


Figure 2: RevTrace architecture (top) and the DGFT commit timeline (bottom). Provisional display and state write are separate permissions; only a certificate-bearing transaction whose parent version and frontier checks pass may advance committed state.

during closure and measurement repairs are documented in Appendix A. All evaluated numbers are bound to code commit, contract digest, and result hashes in the release. The source code, frozen audited artifacts, and analysis scripts are publicly available in the RevTrace repository (`https://github.com/zhouz4884-coder/RevTrace`).

The independent scale-out contract uses one RHEL 7.6 node with eight Tesla V100 PCIe GPUs (32 GB each), PyTorch 2.4.1, CUDA 12.1, and the Transformers-eager backend. It runs replicas rather than distributed model parallelism: up to four proposal workers and four authority workers share a versioned commit queue. Input audio is deterministically resampled to 16 kHz. This contract has its own environment fingerprint, raw cell records, worker diagnostics, and result hashes; its absolute timing is never pooled with the RTX 3090/vLLM contract.

6. Evaluation

6.1. Experimental setup and claim contracts

Table 2 fixes, per family and per model, the proposal path, authority path, workload contract, quality reference, and the claims each contract supports; no number is ever pooled across these contracts. Hardware: two RTX 3090 GPUs. Workloads: the shared bilingual 30 s 16-session contract (c16, 480 s audio) for the same-workload ladder, and the per-family production contracts for cross-architecture evidence. H7 uses a separate RHEL 7.6 single-node contract with eight Tesla V100 PCIe 32 GB GPUs and the Transformers-eager backend. Its normal-load, scale-out, and bottleneck-attribution values are reported independently and are never pooled with RTX 3090/vLLM measurements.

6.2. H1: state divergence propagates through the feedback edge

Prediction (Remark 2.2): under identical future inputs, batch assignments, and arithmetic realization, setting $\beta_t \equiv 0$ by removing the feedback edge eliminates propagation of the isolated batch-created state-entry divergence. *Experiment.* On the Whisper family we enumerated the 21 legal state pairs and intervened causally: the identical perturbation was applied with the feedback edge present and with it removed. *Result.* With feedback, 8 of 21 legal pairs propagate the divergence; without feedback, 0 of 21. On the 6 persistent pairs the measured tail RAF (Definition 2.6) reaches 153. *Verdict.* The prediction is confirmed: the feedback edge is the propagation channel, and contamination is therefore a state-authority problem, not a display problem.

6.3. H2: transaction invariants exclude uncertified ancestry

Prediction (Theorems 3.1–3.5): if the runtime enforces Definitions 3.2–3.3, no install violation, fork, gap, or overlap occurs in any completed trace. *Experiment.* Exhaustive online validation of every transaction install attempt, reported by scope, plus a deterministic protocol-state fuzzer. (i) DGFT online transaction trace: 320 Qwen revision events (R10 cross-adapter matrix, four DGFT cells) and 1,048 Whisper commits (R9.E, 16 cells) and terminal frontier closure. (ii) SG cells on the identical runtime and workload: the two SG ablation cells contribute 240 validated events each. (iii) Protocol fuzzing: 250,000 commit attempts over 64 sessions, cycling through 24 invalid/replay mutation classes (cross-session/epoch/policy, certificate-field tampering, stale parent, gap/overlap, duplicate and replay) and a valid class. This CPU test exercises the commit state machine; it is not a GPU numerical or population failure-rate experiment. *Result.* In both scopes: zero

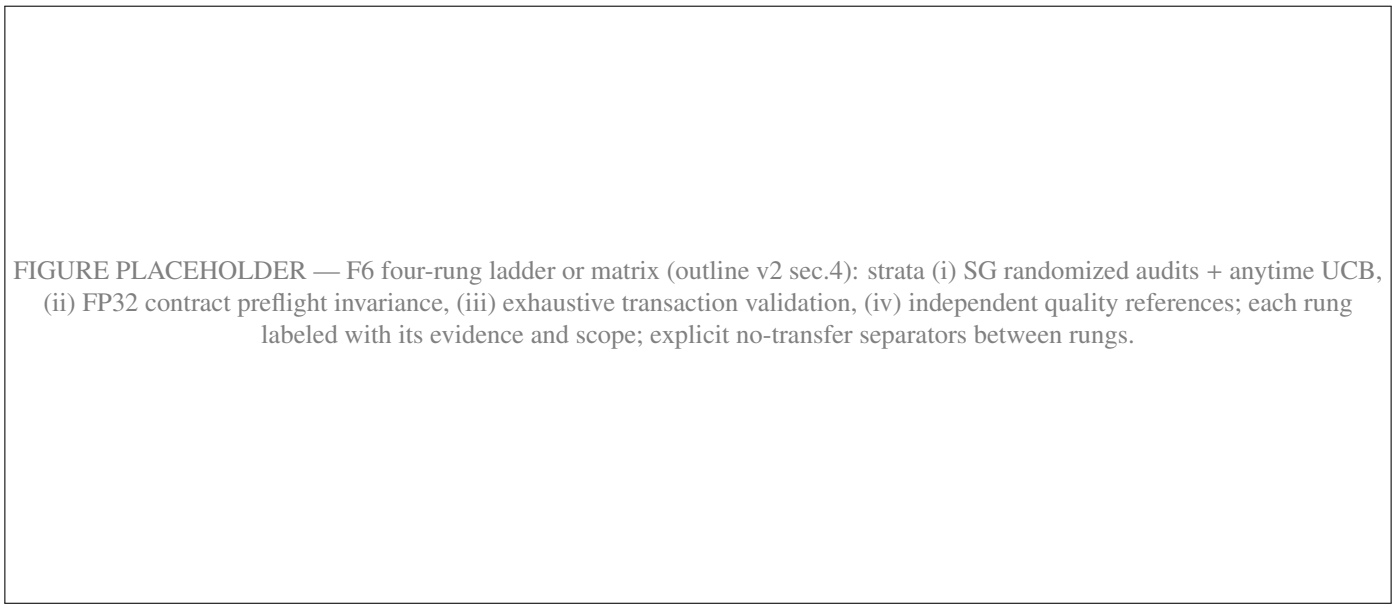


Figure 3: Stratified, contract-scoped certification. Each rung yields exactly one scope-limited guarantee; no rung's evidence authorizes another's installation decisions.

transaction-install violations, zero output/state violations and zero stale or replayed results observed; all sessions closed with monotone terminal frontiers. The evaluated production traces contained no stale install attempt, so they validate accepted-history integrity rather than fault-injected stale-rejection coverage. Batch-repeat and batch-vs-singleton zeros belong to the separate FP32 preflight scope (Section 6.6(ii)) and are not pooled with the online trace evidence. The fuzzer accepted all 10,000 valid commits, rejected all 240,000 mutated/replayed attempts, and recorded zero invalid accepts or state-invariant violations. Its scope excludes Byzantine trusted components, wall-clock expiry and GPU worker crashes. *Verdict.* The invariants hold exhaustively on the evaluated traces. We report this as exhaustive-trace protocol validation; it is not a failure-probability-zero claim for unobserved workloads.

6.4. H3: authority-path configuration shapes the performance frontier

Hypotheses (Proposition 3.8, P1–P4): the evaluated coarse authority path can beat SG on throughput; SG can show terminal-burst tails even at 8–10% verification; Hybrid may pay for quality in performance and energy; and the ordering may reverse across models.

Experiment. Same-workload ladder on the identical 16 bilingual 30 s sessions (c16): SG ($\rho = 1$, FP16 singleton authority, verification fractions 8.3%/9.6%), Coarse-DGFT ($\rho = 3$, FP32 batched authority, $r = 0$), Hybrid-DGFT ($\rho = 3$, $r > 0$), on Qwen3-ASR 0.6B and 1.7B. This is a complete-configuration ablation: authority precision and execution mode change together with granularity. It therefore does not isolate a precision-independent causal effect of ρ. *Result* (Table 3, Figure 4). Direct service wall times are 14.905/16.285 s for SG and 10.759/12.763 s for Coarse-DGFT, yielding the pairwise throughput ratios 1.39× (0.6B) and 1.28× (1.7B) and wall-time reductions of 27.8% and 21.6%. The separately normalized headline speedups in Table 3 use each path's frozen all-singleton reference and must not be divided to recover these pairwise ratios. SG misses its 2 s *certified state-entry* deadline at the terminal burst (miss rates 3.33%/4.17%; P99 2.735/2.964 s). Coarse/Hybrid provide provisional display P99 $\leq$ 1.018 s, while their segment-commit P99 is 2.582–4.437 s. We report these as different latency contracts and do not transfer a pass/fail label between them. Hybrid holds throughput near SG (1.03× on 0.6B) at lower edit rate (2.15% vs 3.43% P95) but costs energy (+16.8% vs SG on 0.6B; +54.2% on 1.7B, where Hybrid also loses throughput, 0.85×). The direct energy comparison against SG reverses between models: Coarse saves 4.4% on 0.6B but consumes 15.1% more on 1.7B. *Verdict.* The observations are consistent with P1–P4, including the model-dependent reversal of the energy ordering, but support a claim about the measured authority-path configurations rather than granularity alone. The SG configuration remains the same-granularity certification point (Corollary 3.6), not a dominated baseline.

6.5. H4: one invariant holds across adapters

Prediction (Section 4.1): the protocol needs only projection, authority, and certificate interfaces, so one runtime serves structurally different revision semantics. *Experiment and result.* On Qwen3-ASR (rollback, holdback, growing buffer), the R10 production matrix reaches 6.88× wall-clock and 3.44× GPU-normalized speedup with 59.8% energy reduction (0.6B, Coarse-DGFT), and 6.06×/3.03× with 51.0% energy reduction (1.7B), at P95 mixed-unit edit rates $\leq$ 3.43% against an independent FP16 singleton reference and display P99 $\leq$ 1.018 s. On Whisper (previous-text conditioning), the identical runtime passes 16/16 production cells with median wall-clock speedup 2.841× (2.633–3.849 across cells), 1.421× GPU-normalized,

Table 2: Claim contracts. Every number in this paper is bound to one row; nothing is pooled across rows. Authority paths and quality references are per-contract by design (Section 6.1).

Family	Model	Proposal path	Authority path	Workload contract	Quality reference	Main claim
Qwen3-ASR	0.6B / 1.7B	fine FP16 batched proposer	FP32 batched (Coarse/Hybrid, $\rho = 3$); FP16 singleton (SG, $\rho = 1$)	shared c16: 16 bilingual 30 s sessions (480 s audio), online causal arrivals, 2×RTX 3090	mixed-unit edit rate vs independent fresh FP16 singleton replay	throughput, energy, latency; transaction safety; bounded edit rate
Qwen3-ASR (scale-out)	0.6B / 1.7B	FP16 batched proposer replicas	FP16/FP32 authority replicas under the V100 matrix contract	c16–c128 bilingual 30 s sessions, one 8×V100 PCIe node, Transformers-eager	per-cell independent singleton replay where applicable	normal-load speedup, bottleneck attribution, single-node SLO capacity and replica scale efficiency
Whisper	tiny / base / small.en	fine FP16 batched proposer	per-cell authority of the R9.E production contract	R9.E production cells (16 cells, per model/arrival group), stable-output hash equivalence	paired WER delta vs unprotected feedback (dev + untouched split)	production wall, GPU, energy; WER improvement; certificate safety

Table 3: Same-workload ladder on the identical 16 bilingual 30 s sessions (c16, 480 s audio; 2×RTX 3090). ρ: granularity ratio; q/r: SG verification / Hybrid replay fraction. Energy Δ is measured against SG on the same workload; the energy-reduction fractions of SG against the all-singleton no-serving baseline are 69.1% (0.6B) and 66.1% (1.7B). Headline wall speedups use each path's frozen all-singleton reference; direct SG–DGFT comparisons use the absolute service wall time. Commit P99 is the certified state-entry/segment-commit scope; Display P99 is the provisional display scope, so no pass/fail label transfers between columns. The 2 s miss rate is defined only for the SG state-entry contract.

Config	Model	ρ	q/r	Service wall (s)	Headline wall	GPU-norm	E (J)	ΔE vs SG	Commit P99 (s)	Display P99 (s)	SG 2 s state miss	P95 edit
SG	0.6B	1	q=8.3%	14.905	5.10×	2.55×	4249.4	—	2.735	n/a	3.33%	0.00%
SG	1.7B	1	q=9.6%	16.285	4.68×	2.34×	5215.7	—	2.964	n/a	4.17%	0.00%
Coarse-DGFT	0.6B	3	r=0	10.759	6.88×	3.44×	4062.2	−4.4%	2.582	0.826	n/a	3.43%
Coarse-DGFT	1.7B	3	r=0	12.763	6.06×	3.03×	6001.9	+15.1%	3.219	1.018	n/a	1.79%
Hybrid-DGFT	0.6B	3	r=5.0%	14.428	5.19×	2.60×	4961.6	+16.8%	3.853	0.782	n/a	2.15%
Hybrid-DGFT	1.7B	3	r=8.8%	19.220	3.92×	1.96×	8045.0	+54.2%	4.437	0.990	n/a	1.79%

31.13% energy reduction, display P99 ≤ 1.492 s, and 1,048 certificates with zero violations. The 16 cells are explicit rather than pooled: Whisper base.en uses five synchronous and five Poisson seeds, while tiny.en and small.en each use three synchronous seeds. *Verdict.* Confirmed: the invariant transfers across two architecture families through adapters; performance magnitudes are family- and model-dependent and are reported per contract.

6.6. H5: each certification layer guarantees only its scope

Prediction (R4, Section 4.4): evidence strata are non-transferable; each yields a bounded statement about its own scope. *Result, by stratum.* (i) SG numerical risk: randomized audits of the SG authority path (≈30k audits, checkpoints 20 s/120 s) give anytime-valid UCBs of 0.480%/0.478%; this bound belongs to the SG adapter path only and is not transferred to DGFT paths [17]. (ii) Authority implementation: the contract-scoped FP32 preflight records zero batch-repeat and zero batch-vs-singleton mismatches on the frozen Qwen contract (10.7–11.1 s, excluded from service timing). (iii) Transaction integrity: H2's exhaustive validation (zero observed violations on 320 DGFT Qwen events, 480 SG Qwen events, and 1,048 Whisper certified commits; no stale attempt occurred in these traces). (iv) Semantic quality, per family and never pooled: Qwen mixed-unit edit rate against the independent FP16 singleton reference has P95 ≤ 3.43% (worst 5.41%, zero catastrophic cases; exact sessions rise from 11/16 to 13/16 on 0.6B Hybrid and from 9/16 to 10/16 on 1.7B); Whisper paired WER deltas against unprotected feedback are all negative, −12.23/−18.21/−25.27 points on dev for tiny/base/small.en (bootstrap uppers −10.48/−16.17/−22.15) and −8.26/−11.03/−13.76 on the untouched cohort (40 long sessions per model, deterministically assembled from 214 previously unused source utterances; uppers −5.15/−7.51/−9.09) (Figure 6). On the independent 64-utterance-per-model-language closure, deployed Qwen records English WER 2.15%/2.30% (0.6B/1.7B), Chinese CER 2.71%/2.16%, terminal exactness 256/256 against singleton authority, zero catastrophic outliers, and partial-revision edit rates 24.7–31.8%.

Verdict. Each layer produces exactly one scope-limited guarantee; no stratum's evidence is used to authorize another's installation decisions.

6.7. H6: faithful revision adaptations expose the missing contract

Published systems target different semantic objects, so their headline numbers are not an apples-to-apples leaderboard. We

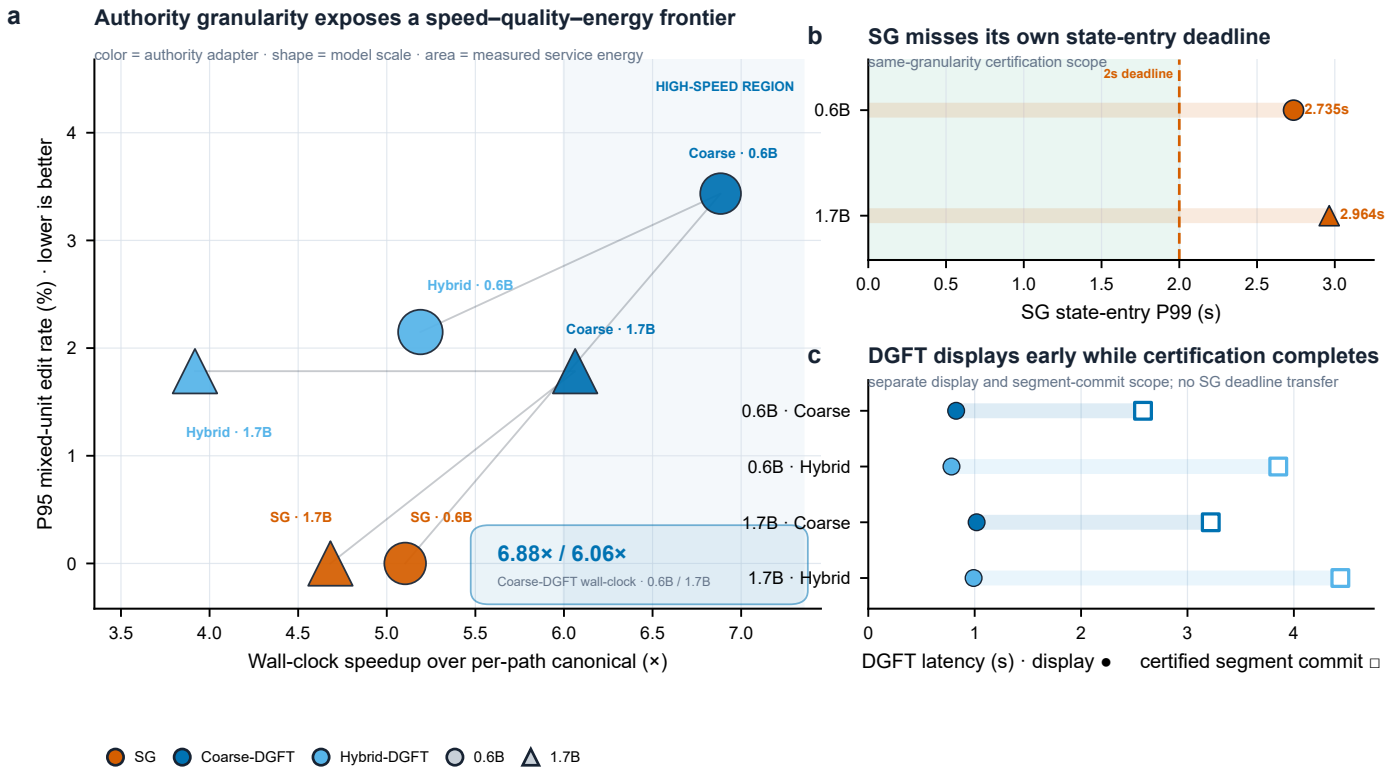


Figure 4: Same-workload operating points (c16). Coarse-DGFT is the throughput point, Hybrid the quality-conservative point, and SG the same-granularity certification point; latency scopes are never mixed on one axis.

Table 4: Cross-family production evidence, per contract. Qwen energy reduction is against the all-singleton FP16 no-serving baseline of the R10 contract; Whisper figures are medians/ranges across the 16 R9.E cells. Violation counts are exhaustive observations over the evaluated traces, not a mathematical claim of an error-free implementation.

Family	Operating point	Wall	GPU-norm	Energy	P95 edit / WER	Display P99	Observed install violations
Qwen3-ASR-0.6B	Coarse-DGFT	6.88×	3.44×	−59.8%	3.43%	0.826 s	0 / 80
Qwen3-ASR-0.6B	Hybrid-DGFT	5.19×	2.60×	−51.3%	2.15%	0.782 s	0 / 80
Qwen3-ASR-1.7B	Coarse-DGFT	6.06×	3.03×	−51.0%	1.79%	1.018 s	0 / 80
Qwen3-ASR-1.7B	Hybrid-DGFT	3.92×	1.96×	−33.2%	1.79%	0.990 s	0 / 80
Whisper (16 cells)	production DGFT	2.841× median (2.633–3.849)	1.421×	−31.13%	paired WER Δ (Table 5)	≤ 1.492 s	0 / 1,048

instead run faithful *revision-level adaptations* on the same frozen overloaded c32 trace (Table 6). A MarginGate-inspired fixed threshold verifies 3.02% of events but misses the known state-changing flip; an owner/version-only baseline rejects stale work but still accepts that legal current-owner flip. An LLM-42-style adaptation verifies every revision and eliminates the observed state error at 1.565 events/s. RevTrace's scoped repair verifies 5.05%, eliminates the error, and reaches 3.046 events/s (1.95× the always-verify adaptation). This overloaded trace remains outside the 2 s SLO (P99 6.107 s), so the comparison isolates safety cost rather than serving capacity. These are adaptations, not end-to-end claims about the original text-only systems [19, 13].

6.8. H7: proposer–authority scale-out preserves the SLO until the measured capacity boundary

Prediction (Equation (9)): authority-only scaling cannot remove a proposer bottleneck; balanced proposal/authority replication should increase service capacity until one stage saturates. Coarse FP32 authority should be most expensive at $\rho = 1$ and become amortizable as ρ increases. *Experiment.* We execute an independent 84-cell matrix on one 8×Tesla-V100-32 GB node: 24 authority microprofiles, 16 normal-load c16 baselines, 32 scale cells over c16/c32/c64/c128 and p1a1/p1a2/p2a2/p4a4, and 12 FP16/FP32 authority-attribution cells at $\rho \in \{1, 3, 5\}$. Each 30 s c16 baseline contains 240 display events; each c64 cell contains 960. The complete raw matrix and worker diagnostics are audited separately from the RTX 3090 contract.

Result. First, all repeated batched V1 executions agree, yet each model accumulates 62 batch–singleton request mismatches across FP16/FP32 and batch sizes 1–32. FP32 is therefore not an unconditional singleton-equivalence certificate on this platform. Second, under the normal-load c16 contract, SG reaches 4.682 × /3.248× wall speedup with 6.67%/10.00% verification and 0% P95 final edit; Hybrid reaches 5.764 × /4.657×

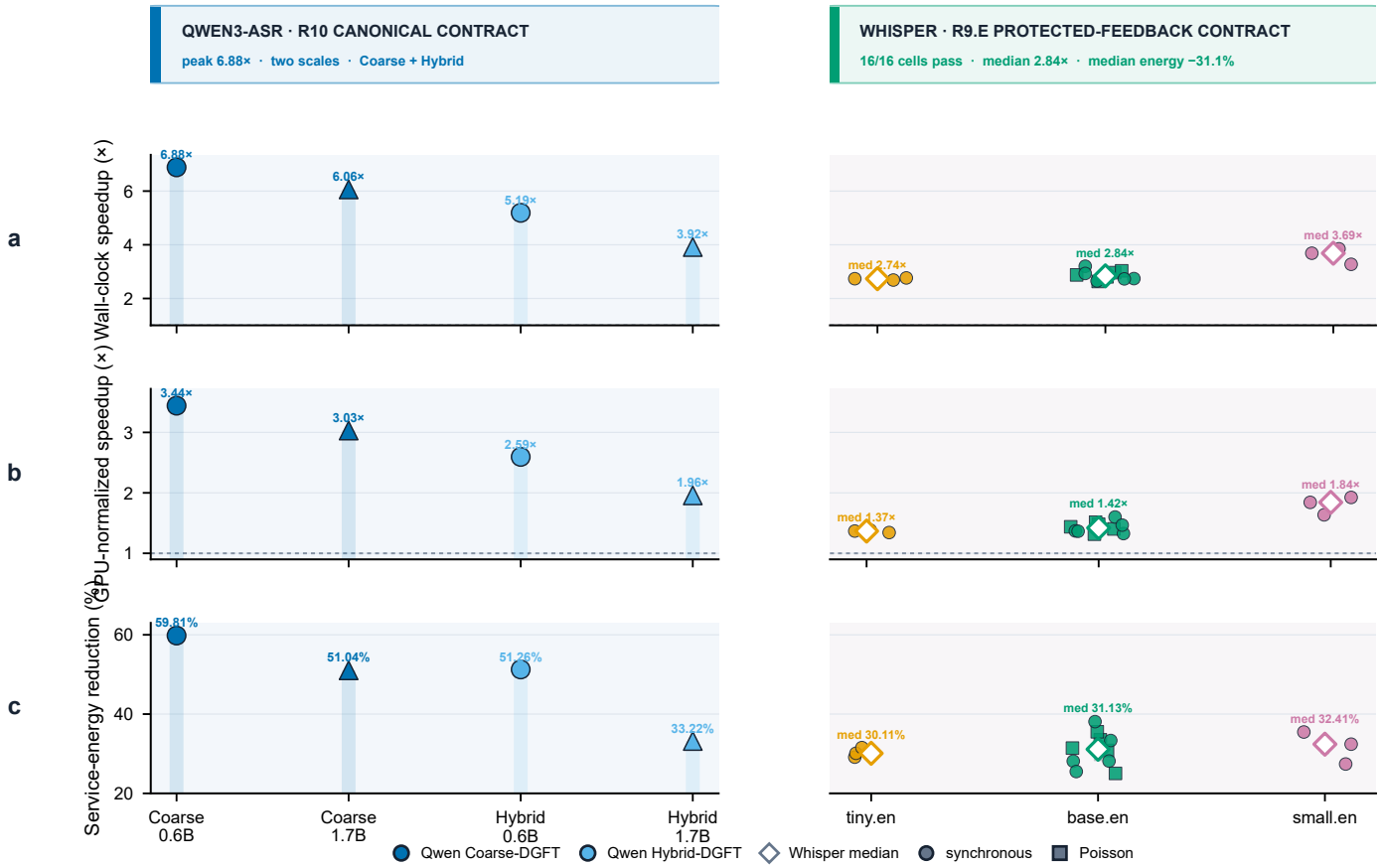


Figure 5: Cross-family production evidence under per-contract baselines. One DGFT abstraction holds across both architecture families, but gain magnitudes are authority-workload- and model-dependent; the Qwen and Whisper panels are different contracts and are never compared horizontally.

(Table 7). Third, p4a4 at c64 sustains 147.138 × /114.488× audio real-time throughput with display P99 1.115/1.471 s, zero misses among 960 display events per model, and nominal scale efficiency 91.1%/100.7% relative to p1a1 at the same load. At c128, throughput saturates at 148.930 × /111.133× while P99 rises to 2.266/2.799 s; c64 is therefore the measured 2 s-SLO capacity for both models. Finally, at $\rho = 1$ the FP32 authority/proposer service ratios are 2.216/2.669 and throughput falls to 18.603 × /12.294×; at $\rho = 5$ these ratios fall below one (0.759/0.841), restoring 40.630 × /33.909× at c16. Across the matrix we observe no transaction-install, uncertified-ancestry, or directional-omission violation.

Verdict. The scale and attribution results support the two-stage cost model: balanced replication, rather than authority replicas alone, expands capacity, and coarse authority becomes economical only after sufficient amortization. The result closes the single-node high-concurrency gap but does not establish multi-node scaling; c128 is reported as a failed SLO boundary.

7. Discussion and Boundaries

FP32 authority is a canonical path, not a repair.. The FP32 batched authority defines a new canonical reference for the contract; it is not a bitwise repair of FP16 arithmetic. FP16 remains the proposer and SG-authority precision. Quality estimands are defined against the FP16 singleton reference (Qwen) and against unprotected feedback (Whisper); neither claims bitwise equivalence across precisions.

Risk evidence does not transfer.. The SG-path UCB of 0.480% bounds the numerical risk of the SG adapter's randomized audits. It is never applied to Coarse/Hybrid paths, whose safety rests on Theorems 3.1–3.5 and exhaustive transaction validation instead.

The authority plane is trusted and versioned.. The current contract assumes a non-Byzantine authority coordinator, atomic certificate consumption and version comparison, immutable audio bytes during a transaction, and collision-resistant digests. A certificate binds the parent version, audio and projection digests, and the complete canonical-path configuration; it is not a standalone proof against a compromised authority. Our production traces contain no stale-install attempt. Their zero observed install violations therefore validate accepted-history integrity, whereas the stale-result rejection rule is supported by the protocol argument rather than a fault-injection denominator.

Quality estimands differ by family.. Qwen edit rates measure distance to an independent canonical reference; Whisper WER deltas measure improvement over unprotected feedback. The

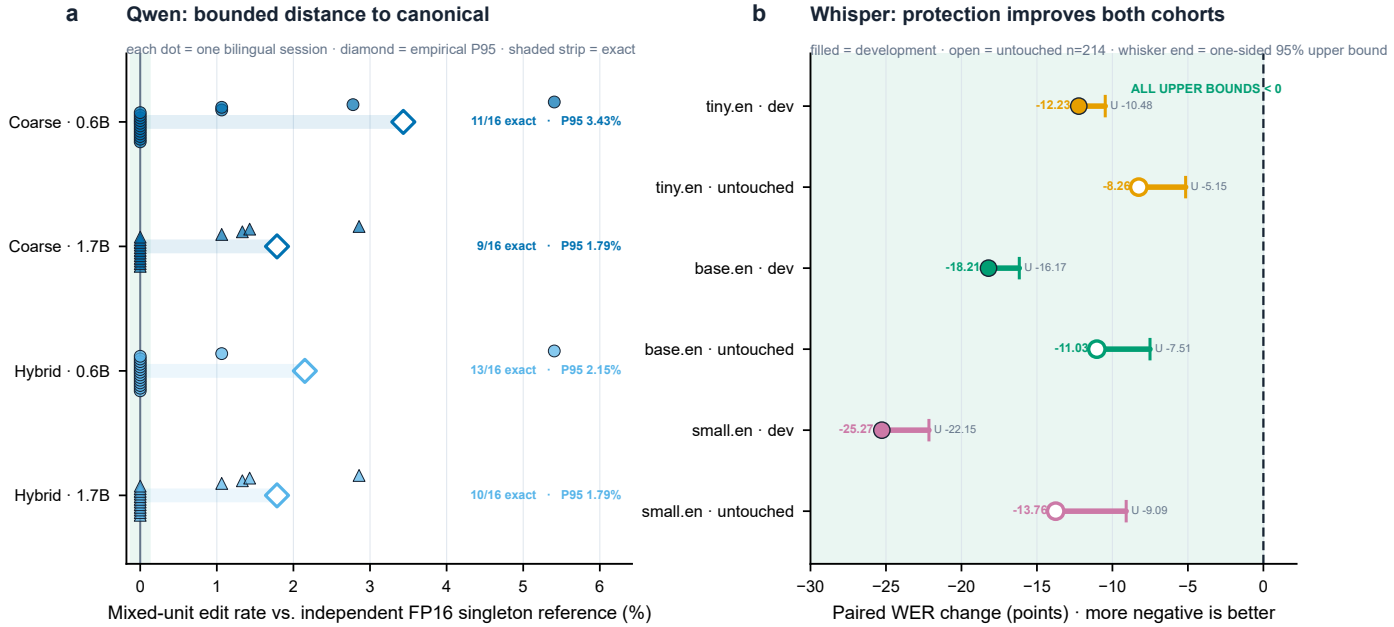


Figure 6: Per-family quality contracts. Qwen measures distance to an independent canonical reference; Whisper measures improvement over unprotected feedback. Open markers: untouched cohort, $n = 40$ long sessions per model, deterministically assembled from 214 previously unused source utterances (bootstrap unit: long session).

two answer different questions and are never pooled into a single quality number.

Virtual arrival versus wall clock.. Production cells schedule virtual causal arrivals; service wall-clock and energy meters measure real GPU work. Latency columns are labeled by semantics (display, state-entry, or segment-commit) wherever they appear. Consequently, the SG certified state-entry deadline and the DGFT provisional display/segment-commit measurements are reported separately; the manuscript does not transfer an SLO pass label between them.

Why coarse authority improves throughput.. The same-workload ablation supports the cost decomposition rather than a granularity-only explanation. SG repeatedly invokes an FP16 singleton authority and accumulates a synchronized terminal burst; Coarse amortizes authority work through an FP32 batch, while Hybrid pays selective replay to recover a more conservative quality point. This explains why Coarse has the shortest absolute wall time, why Hybrid is not uniformly dominant, and why the ordering may change with model-specific FP16/FP32 cost. Unlike token-level verified speculation such as LLM-42 [13], the saved work here is measured at a revision-state authorization boundary; unlike stateful cache reuse [20, 21], the protected object is the transition that creates reusable state.

The cost equations are a decomposition, not a fitted oracle.. Equations (7)–(9) identify the complete authority-path terms and yield a sufficient operating condition; they do not claim a workload-independent numerical predictor. H3 therefore compares complete configurations on one frozen workload and reports the model-dependent ordering reversal. Risk-bound sensitivity is evaluated separately and is never converted into fabricated per-setting GPU times.

Preflight amortization.. The one-time contract preflight costs 10.7–11.1 s. With per-session steady-state savings $\Delta T = T_{\text{baseline}} - T_{\text{DGFT}}$, we define the break-even point as $N = \lceil T_{\text{preflight}}/\Delta T \rceil$ sessions. We report the one-time cost and steady-state results separately, and exclude the preflight from all service timing.

Not every ASR system needs DGFT.. The original Voxtral report studies multimodal audio understanding and long-context transcription [22]; Voxtral Realtime instead uses a natively streaming delayed-stream design [23]. Such append-only semantics expose no rollback-style feedback write path, so the contamination channel of Section 2.4 is absent; at low concurrency (our c8 capacity probe) batching pressure is small. DGFT targets the revision-generative operating regime where feedback writes and batching coexist.

Scale boundary.. Under the stated trust and atomicity assumptions, safety is independent of adapter and granularity, but performance remains hardware- and backend-specific. The RTX 3090 results establish cross-architecture mechanism portability over five checkpoints. The independent 8×V100 experiment adds single-node replica scaling and identifies c64 as the largest measured point satisfying the 2 s display SLO; c128 saturates and fails that SLO. Neither experiment establishes multi-node scalability or a data-center production claim. The V100 result also shows that even FP32 batching need not match singleton output, so its authority contract is platform-scoped rather than assumed from precision alone. Higher-end accelerators and distributed authority ledgers may change the

Table 5: Per-family quality contracts. The two zones answer different questions and are never pooled: Qwen measures distance to an independent canonical reference; Whisper measures improvement over unprotected feedback. Bootstrap upper bounds are one-sided 95%; the bootstrap unit is the long session. The untouched cohort is deterministically assembled from 214 previously unused source utterances into 40 long sessions per model, not 214 independent sessions.

Zone	Quantity	Value	Scope
Qwen3-ASR — mixed-unit edit rate vs independent FP16 singleton reference			
	P95 edit rate (worst cell)	$\leq 3.43\%$ (worst 5.41%)	R10 cells, c16
	catastrophic cases	0	R10 cells
	exact sessions, 0.6B Hybrid	$11/16 \rightarrow 13/16$	c16
	exact sessions, 1.7B Hybrid	$9/16 \rightarrow 10/16$	c16
	English WER, 0.6B / 1.7B	2.15% / 2.30%	64 utterances/model
	Chinese CER, 0.6B / 1.7B	2.71% / 2.16%	64 utterances/model
	terminal exact / catastrophic	256/256 / 0	singleton authority
	partial-revision edit rate	24.7–31.8%	four model-language cells
Whisper — paired WER delta vs unprotected feedback (points; negative = better)			
	dev, tiny / base / small.en	-12.23 / -18.21 / -25.27	uppers -10.48 / -16.17 / -22.15
	untouched (n=40 long sessions/model)	-8.26 / -11.03 / -13.76	uppers -5.15 / -7.51 / -9.09

Table 6: Faithful revision-level adaptations on the same frozen overloaded c32 trace. "State err." counts observed state-entry disagreement. Goodput is reported only where measured under the adaptation. The trace is outside the 2 s SLO and therefore diagnoses safety cost, not production capacity.

Mechanism	Verify fraction	State-write authority	Stale fence	State err.	Goodput	Boundary
MarginGate-inspired	3.02%	no	no	1	–	misses known 0.109375-margin flip
Owner/version only	–	no	yes	1	–	legal current-owner flip accepted
LLM-42-style revision	100%	canonical each revision	implicit	0	1.565 evt/s	text-only source adapted
RevTrace scoped repair	5.05%	certificate bound	yes	0	3.046 evt/s	P99 6.107 s; overload

cost ordering without changing the state-write authorization problem.

8. Related Work

Streaming ASR and confirmation policies.. Partial-hypothesis stability has long been recognized as distinct from final recognition quality [24, 25]. Modern streaming decoders therefore manage revision at the display layer: transducers emit and revise hypotheses as audio arrives [26], while Whisper-based streaming maintains confirmed and unconfirmed prefixes and commits text after repeated decoding passes agree [2, 1]; chunk-based adaptations additionally detect truncation [27]. These policies reduce visible flicker. RevTrace addresses a different boundary: whether a generated projection is authorized to become a parent of future model state. A display-confirmed prefix need not be certified against an independent authority path, and a provisional display need not be allowed to update feedback state.

Speculative execution and numerical verification.. Speculative decoding verifies draft tokens against a target model so that acceleration preserves the target distribution [11, 12]; Medusa, EAGLE, and SpecInfer increase the accepted work per verification step [28, 29, 30]. More directly, LLM-42 combines a fast nondeterministic path with a fixed arithmetic verification path and rollback [13]. RevTrace shares the propose–verify pattern but changes both the verified object and the safety condition: it certifies a revision transition whose projection can condition all later rounds, binds that certificate to parent provenance, and prevents uncertified descendants from entering committed state. Thus token acceptance and revision-state installation are related but non-interchangeable contracts. MarginGate sparsifies LLM-42-style verification using low token margins and repairs the current KV column [19]; its risk signal does not itself define feedback-state authority, certificate binding, or parent-version installation. H6 evaluates a clearly labeled revision-level adaptation rather than attributing those semantics to the original system.

LLM serving and materialized state.. Throughput-oriented serving systems optimize request execution through paged KV-cache management [4], iteration-level scheduling [5], structured-program runtimes [6], chunked prefill [7], phase disaggregation [8], and request migration [9]. Stateful systems further preserve, place, or reuse materialized state: Pensieve and CachedAttention reuse or transport multi-turn KV state [20, 21], Parrot schedules application dataflow with semantic variables [31], Agentix manages cross-agent KV reuse [32], and Mooncake disaggregates KV-cache storage [33]. These mechanisms reason about the identity, residency, transfer, or dependency of already materialized state. They do not

Table 7: Independent single-node V100 results. V2 uses the same c16 bilingual 30 s workload for SG and Hybrid; speedups are against the V100 FP16 all-singleton baseline. V3 reports the SLO-valid p4a4 c64 point; nominal scale efficiency compares p4a4 with p1a1 at c64 and normalizes by the 4× GPU count increase. A value near 100% is not a claim of statistically significant superlinear scaling. V100/Transformers-eager numbers are not pooled with the RTX 3090/vLLM contract.

Model	SG wall	SG display P99	SG verify	SG P95 edit	Hybrid wall	c64 audio xRT	c64 display P99	Scale eff.
Qwen3-ASR-0.6B	4.682×	1.095 s	6.67%	0.00%	5.764×	147.138×	1.115 s	91.1%
Qwen3-ASR-1.7B	3.248×	1.440 s	10.00%	0.00%	4.657×	114.488×	1.471 s	100.7%

generally expose a separate authorization point at which a batch-created, model-generated revision projection must be certified before it becomes future feedback. RevTrace is therefore complementary: it governs state-entry authority before downstream cache placement or reuse.

Revision-generative and self-correcting models.. Language models also revise outputs through explicit application loops such as self-refinement [34] and verbal feedback retained by agents [35]. RevTrace targets the narrower systems case in which generated text is fed back automatically by the serving pipeline. Under this feedback edge, a numerically induced output difference is not merely a different terminal answer: it can change the next request and create an ancestry of dependent revisions.

Numerical variation.. Parallel reductions, kernel selection, and low-precision arithmetic can make mathematically equivalent LLM inference paths numerically non-identical [10, 36]. Reproducible reductions are possible, often with implementation or performance cost [37, 38]. RevTrace does not claim to make the proposer bitwise deterministic. It instead names a canonical authority path, treats proposer divergence as a first-class possibility, and restricts state installation to certificate-bearing transitions. This distinction is important because a small one-round numerical difference can become a durable causal difference only when the serving state machine feeds it forward.

Concurrency control and recovery.. RevTrace borrows validation and version fencing from optimistic concurrency control [14] and provenance tracking from multiversion concurrency control [15]. Cascading invalidation is reminiscent of optimistic simulation rollback in Time Warp [39], while partial external effects motivate the longstanding distinction between rollback and compensating actions in Sagas [40]. The resulting guarantee is deliberately narrower than Herlihy–Wing linearizability [16]: RevTrace proves a single fork-free committed revision history under its certificate and parent-version rules, not a real-time specification for every client invocation and response. The validated object is also unusual—a model-generated state projection rather than a conventional database write.

Anytime-valid statistics.. Confidence sequences and e-value-based tests remain valid under optional stopping [17, 18, 41]. RevTrace uses them only to monitor the randomized audits of the SG adapter. They neither certify individual DGFT transitions nor replace the deterministic transaction invariant; each statistical statement is scoped to its declared adapter, sampling process, and estimand.

Synthesis and gap.. The closest lines therefore solve complementary parts of the problem: streaming confirmation governs display stability, verified speculation governs token acceptance, stateful serving governs reuse, and concurrency control governs conventional writes. To our knowledge, prior work does not jointly define (i) a projection from generated revision output into future feedback state, (ii) a certificate that authorizes that projection against a canonical path, and (iii) parent-version rules that exclude uncertified descendants. RevTrace contributes this intersection and evaluates its cost under two revision-generative architecture families; it does not claim to supersede the numerical, caching, or scheduling mechanisms above.

9. Conclusion

Revision-generative serving turns generated outputs into speculative writes to future model state. This paper formalized the resulting contamination channel and introduced certificate-bearing transactions that separate display authority from state-write authority. Under the stated trust and binding assumptions, the protocol excludes uncertified ancestry, produces a single fork-free committed revision history, and remains safe for any granularity ratio. Its cost is governed by the complete authority path—not granularity alone—and the measurements identify when coarse authority is worthwhile. Across two architecture families and five checkpoints, RevTrace reaches up to 6.88× wall-clock speedup. We observe no output/state or transaction-install violations in 800 Qwen revision events (480 SG and 320 DGFT) and 1,048 Whisper commits; these finite traces support the declared contracts but are not a proof of zero implementation faults. Same-granularity certification is the $\rho = 1$ reduction of the protocol. Risk signals schedule verification but never authorize installation, and quality evidence does not transfer across estimands or canonical paths. On an independent single-node 8×V100 contract, balanced four-proposer/four-authority scaling sustains 147.1 × /114.5× audio real-time throughput for Qwen3-ASR 0.6B/1.7B at c64, with display P99 1.115/1.471 s and nominal scale efficiency 91.1/100.7%. The c128 points fail the 2 s SLO and are retained as the measured capacity boundary. These values extend the evidence to high-concurrency single-node serving without being pooled with the RTX 3090/vLLM contract. Extending the transaction discipline to other feedback-driven revision models—including text-LLM and multimodal settings with stateful generation—is the natural next step.

CRediT authorship contribution statement

Bin Zhou: Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft. **Jiyuan Miao:** Visualization. **Xi Cao:** Supervision, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

OpenAI ChatGPT was used as an auxiliary tool for language editing and LaTeX formatting assistance. It was not used to generate experimental measurements. All technical ideas, implementations, numerical results, claims, figures, and final manuscript decisions were independently reviewed and verified by the authors.

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Data availability

The source code, frozen audited result artifacts, replay manifests, and analysis scripts supporting this study are publicly available in the RevTrace repository at `https://github.com/zhouz4884-coder/RevTrace`.

Appendix A. Rejected realizations and measurement repairs

Three deviations were found and resolved during closure, and are documented with the frozen artifacts. (i) An FP16 singleton coarse-authority candidate was rejected after failing its quality contract; FP32 was adopted as the coarse-authority precision instead (the SG path retains its own FP16 singleton authority under the $\rho = 1$ protocol, Corollary 3.6). (ii) The FP32 KV-cache bound was reduced from 65,536 to 4,096 tokens per session. (iii) A per-row CUDA-synchronization defect in the online submission path was fixed before the R9.E closure. All numbers reported in this paper were re-audited after these repairs and are bound to the release's code commit, contract digest, and result hashes.

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